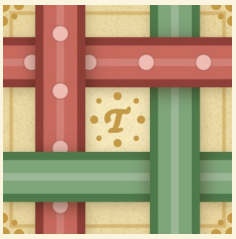
Repeating Knot Cards

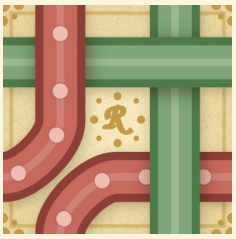
Justin Roughley, 2026

Four cards, nine wallpaper groups

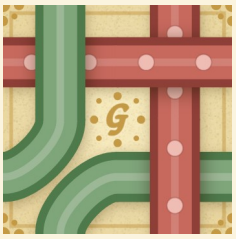
A pack of 52: forty-eight woven cards and four reference cards. Every card is a little patch of Celtic knotwork, and every one can be turned any way: wherever it goes, the ribbons still pass alternately over and under. Lay four cards in a square, repeat the square, and the knot takes on one of nine kinds of symmetry: nine of the seventeen wallpaper groups (en.wikipedia.org/wiki/Wallpaper_group).



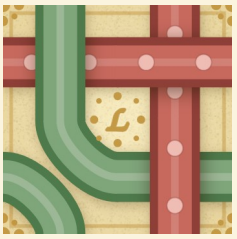
T × 16
four crossings: a trellis



R × 8
two red bends



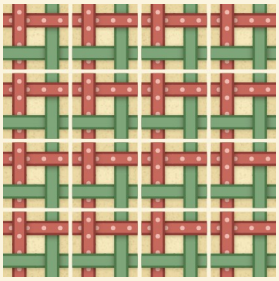
G × 16
two green bends



L × 8
a rounded L and a corner

MAKE A GROUP

1. Lay four cards in a square as one of the recipes below. The letter in the middle of each card shows which way up it is; ↻1 means turn it a quarter turn clockwise, ↻2 a half turn, ↻3 three quarters.
2. Copy that square three times, with the same cards turned the same way, to make a square of sixteen.
3. Check that colours match across every edge, so each strand keeps one colour. Then look for the symmetry.
4. Each group is for the weave: over and under count, colours do not. Beside each recipe, *With colours* gives the group when the colours count too, and *Shape only* the group when over and under are ignored.

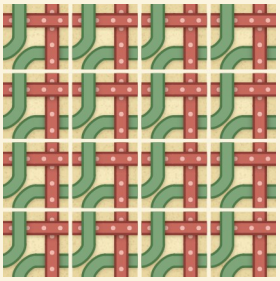


4*2 (p4g)

T · T
T · T

The plain weave: quarter turns and mirrors

With colours ***2222**
Shape only ***442**

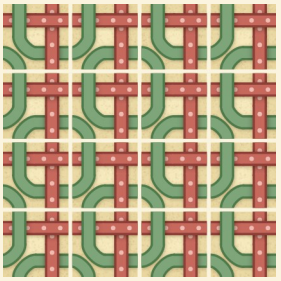


2222 (p2)

G · G
G · G

Half turns only

With colours same
Shape only **2*22**

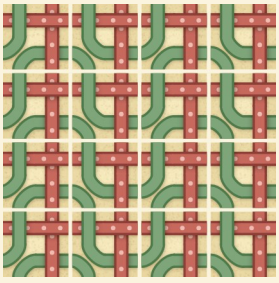


2*22 (cmm)

L · G
G · L

Mirrors in two directions, half turns

With colours same
Shape only ***442**

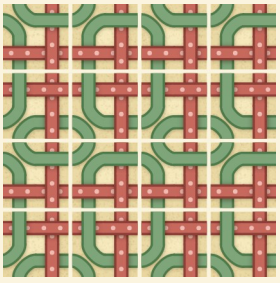


22* (pmg)

G · L
G · L

Parallel mirrors, half turns, glides

With colours same
Shape only same

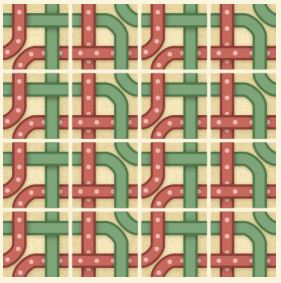


****** (pm)

G ↻1 · L ↻1
G · L

Parallel mirrors only

With colours same
Shape only ***2222**

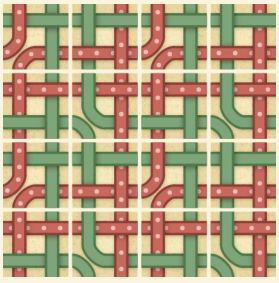


×× (pg)

R · L ↻2
R · L ↻2

Glides only: reflect and slide

With colours **o**
Shape only same

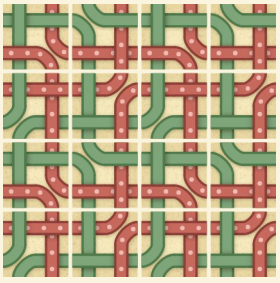


22× (pgg)

R · T ↻2
T · L

Half turns and glides, no mirrors

With colours **o**
Shape only **4*2**

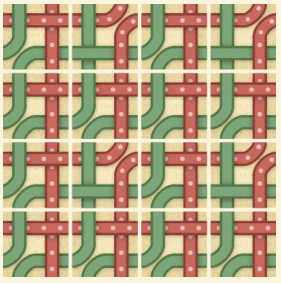


442 (p4)

R ↻3 · G ↻1
G · R ↻2

Quarter turns, no mirrors

With colours **o**
Shape only **4*2**



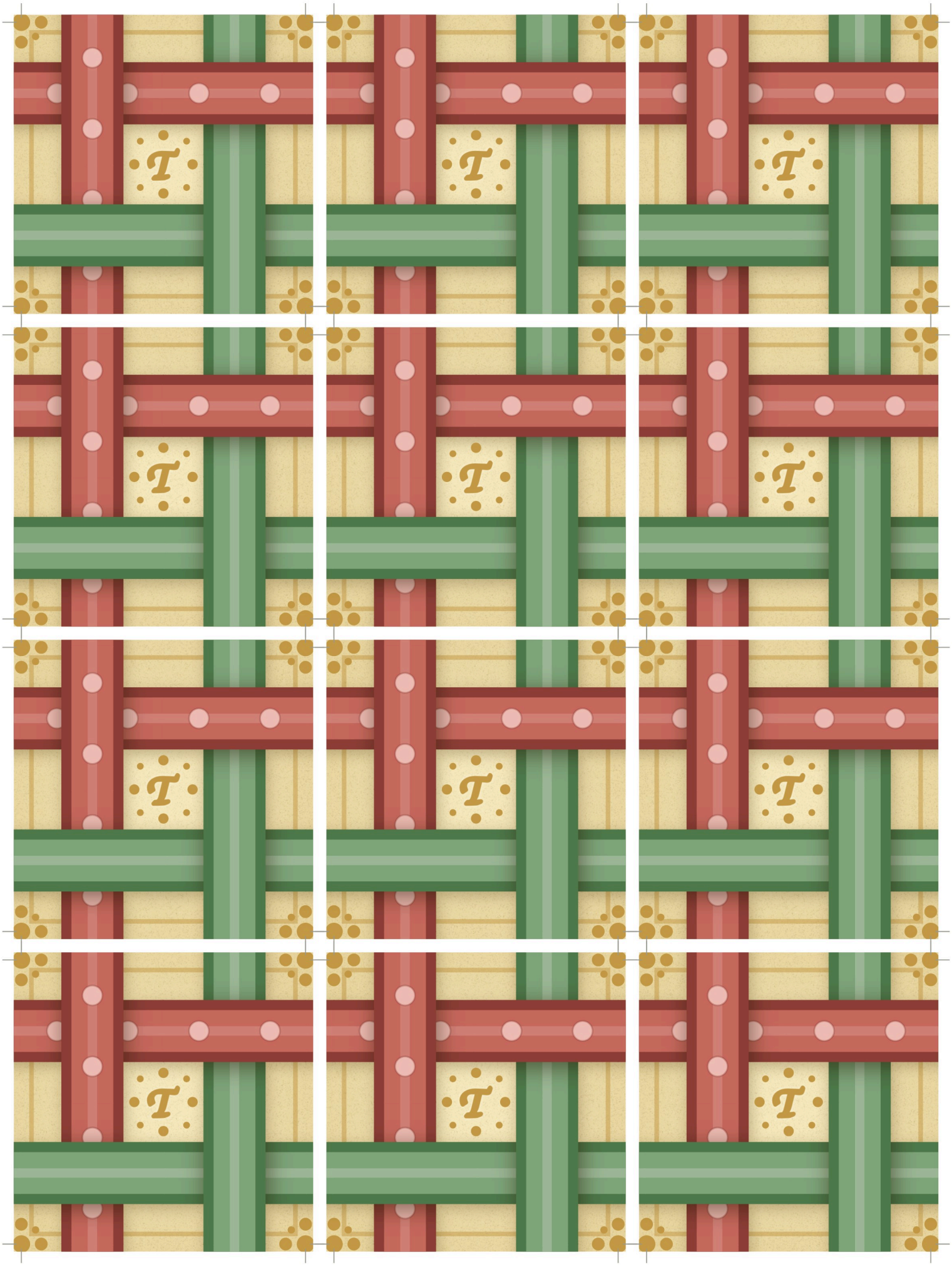
o (p1)

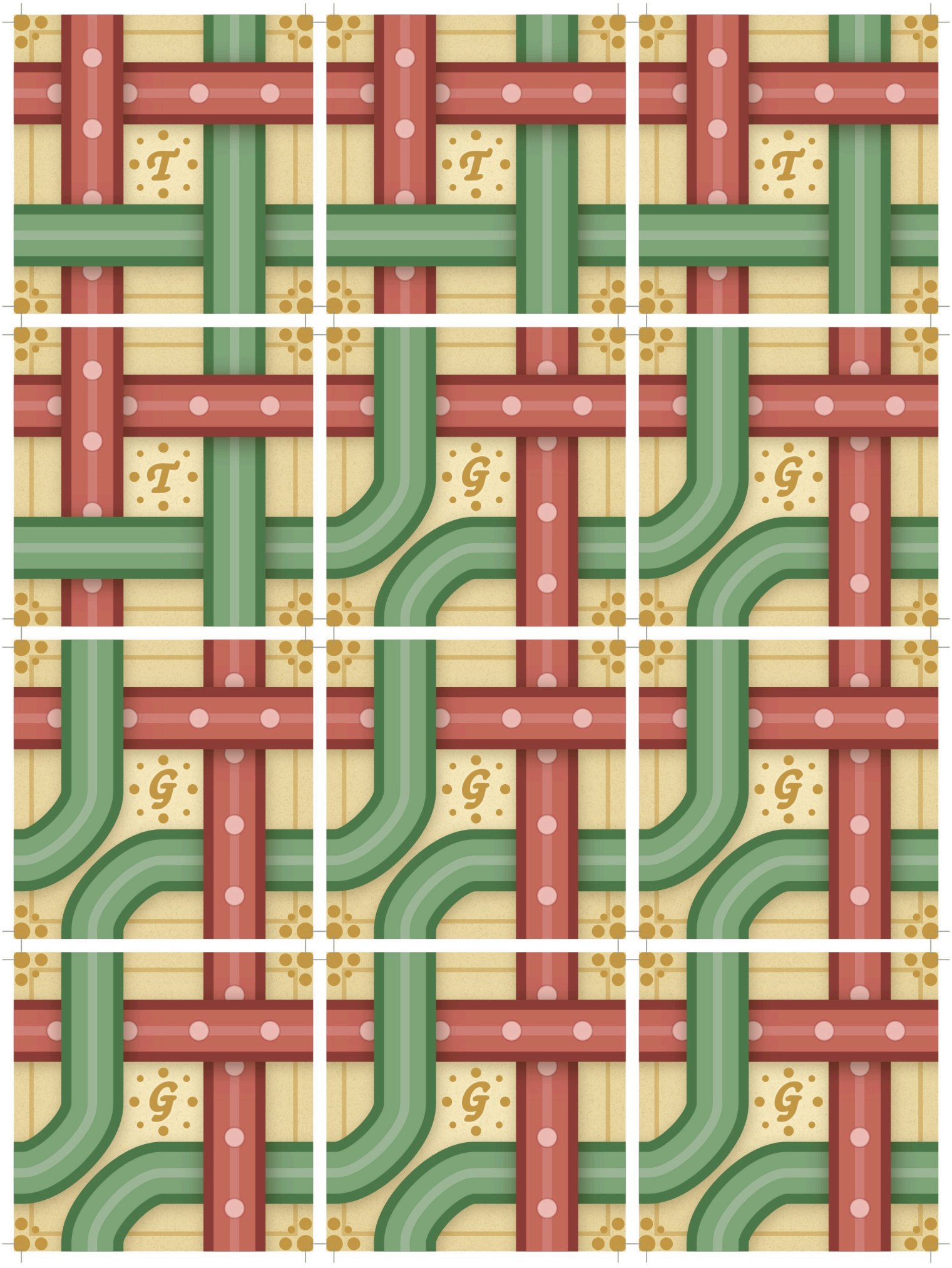
G · R ↻2
G · G

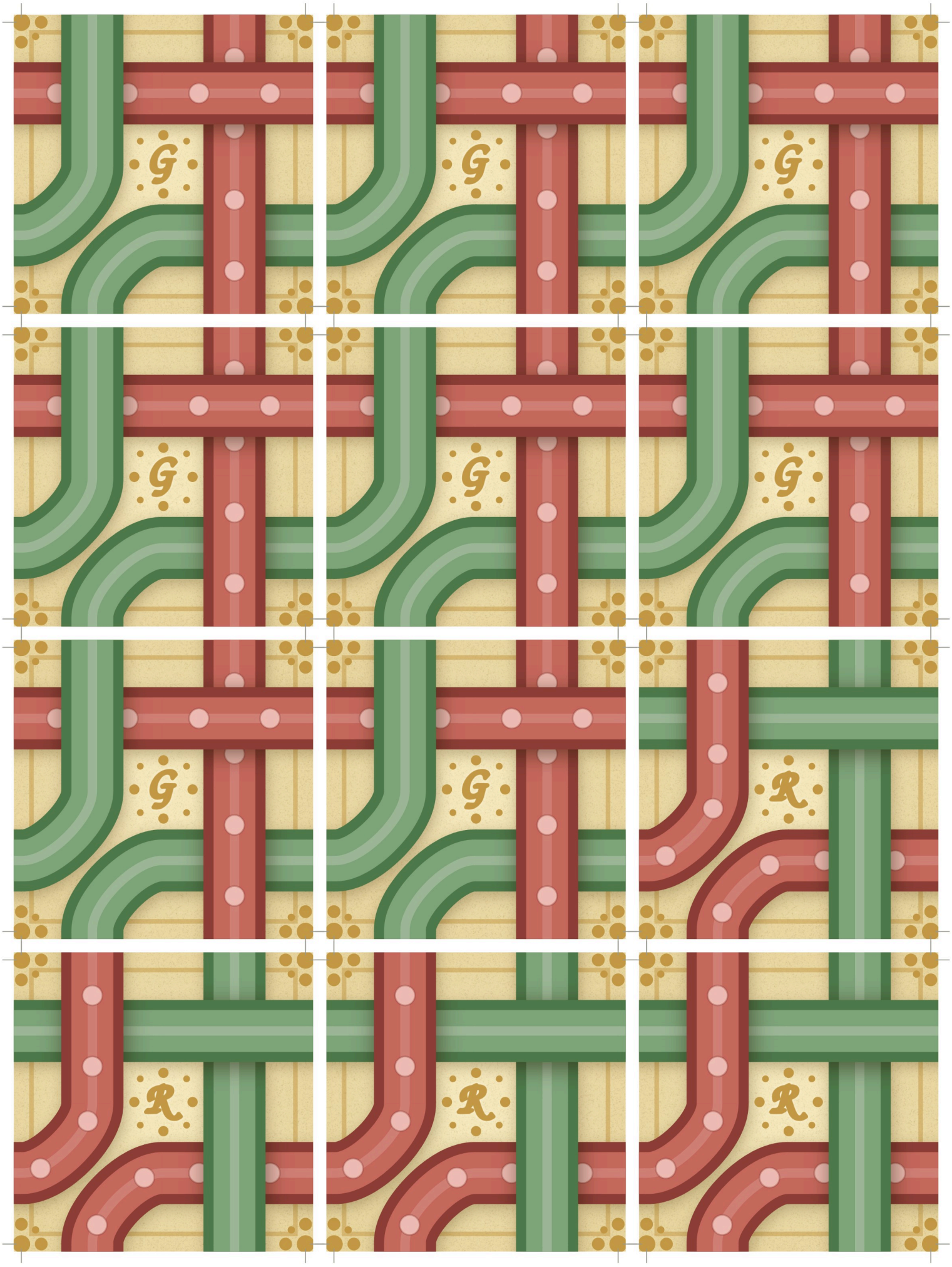
Translations only

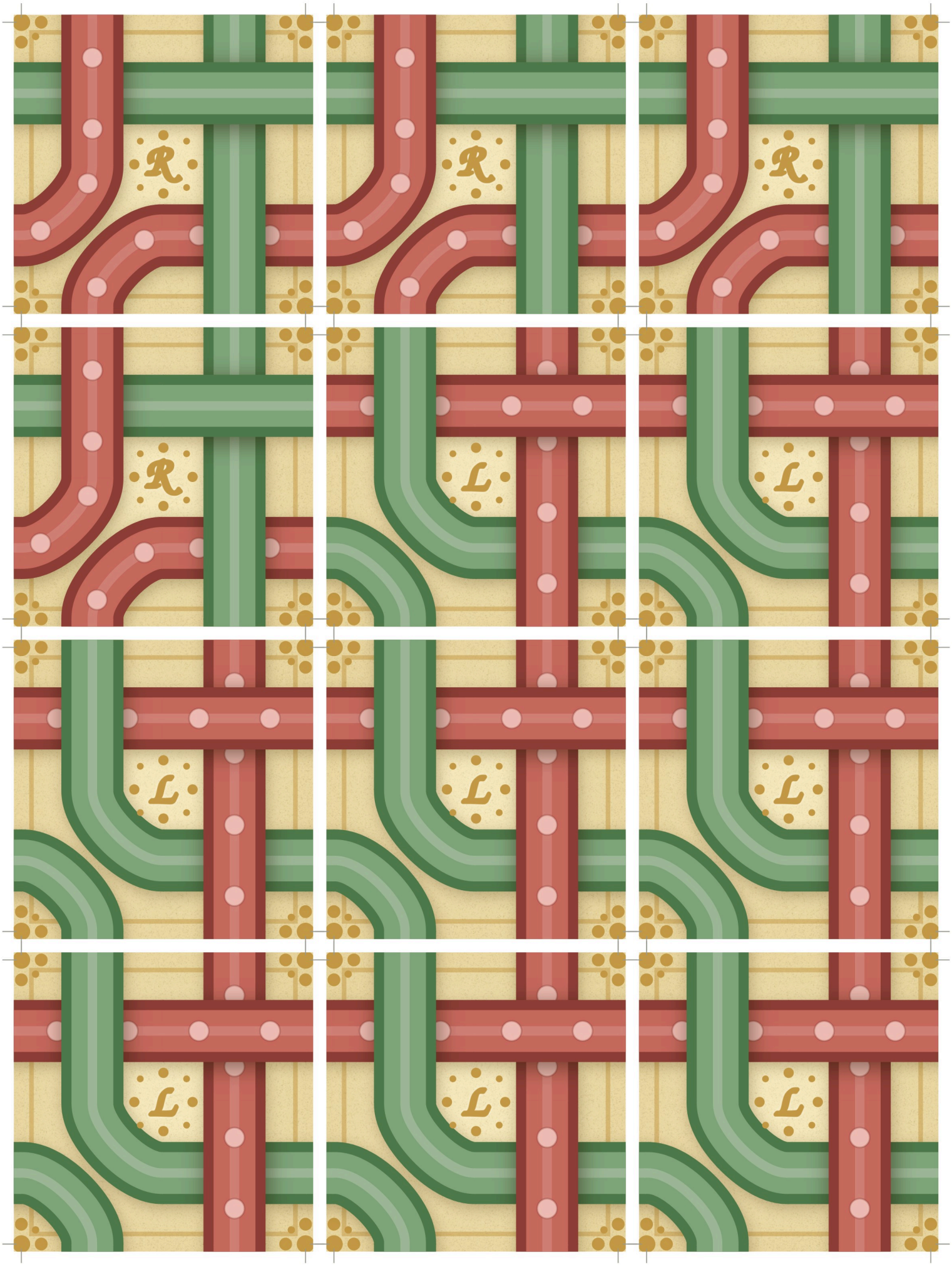
With colours same
Shape only ******

Free to print from https://github.com/Cortado-J/REPEATING_KNOT/tree/main/KNOT_CARDS




T
T
T
T
G
G
G
G
G
G
G
G


G
G
G
G
G
G
G
G
G
R
R
R
R
R


R
R
R
R
L
L
L
L
L
L
L
L

REFERENCE CARDS

Four cards to complete the pack of 52.

Repeating Knot Cards

four cards · nine wallpaper groups

1. Lay four cards in a square.
2. Copy the square to make sixteen.
3. Match ribbon colours across every edge.

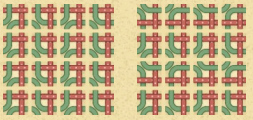
Turn a card any way: it still weaves.



TARGETS I

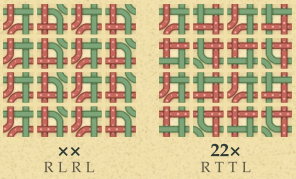


4*2 TTTT **2222** GGGG **2*22** LGGL

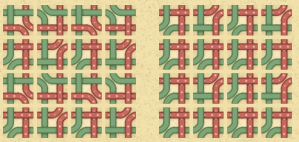


22* GLGL ****** GLGL

TARGETS II



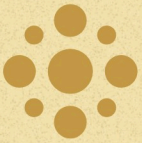
×× RLRL **22×** RTTL



442 RGGR **o** GRGG

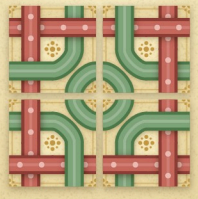
TO WONDER ABOUT

1. Why can't these cards make ***×**, the tenth group a square grid allows?
2. Change one card at a time. Can you visit all nine groups without repeating one?
3. How many strands of each colour does each pattern have?



Repeating Knot Cards

four cards · nine wallpaper groups



A PACK OF 52

Repeating Knot Cards

https://github.com/Cortado-J/REPEATING_KNOT/tree/main/KNOT_CARDS

Justin Roughley